

COGNITIVE AI • OFFLINE •
PRIVATE



Bharosa AI

Personal Multimodal AI That Thinks Offline

Offline Multimodal Conversational AI Assistant using Retrieval-Augmented Generation (RAG)

Problem ID: TIB2625

WHY THIS PROJECT?

- Solves information overload on personal devices
- Delivers true intelligent reasoning over user's own data
- Fully offline with complete privacy protection
- Advanced multimodal RAG architecture optimized for mobile edge devices

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[BHAROSA AI]



Cognitive AI

Privacy First

Multimodal Memory

Edge Optimized

No API • Local Processing • Zero Cloud

Team: **Agile&Empathy**

⚠ The Challenge

Modern users store thousands of documents, images, lectures & notes on their phones — but cannot effectively recall or reason over them. Cloud AI assistants compromise privacy and lack deep understanding.

💡 Our Solution

CogniMem is a fully offline, multimodal cognitive AI system that continuously builds semantic memory from all user files (PDFs, images, audio, video) and delivers intelligent, context-aware reasoning.

KEY CAPABILITIES



Semantic Chunking & Merging



Hybrid Cognitive Classifier



Vector + Metadata Dual-Layer Memory



Temporary Query-Specific Graphs



Cross-Modal Retrieval



Impact: Transforms your smartphone into a true personal cognitive assistant while keeping all data 100% private.

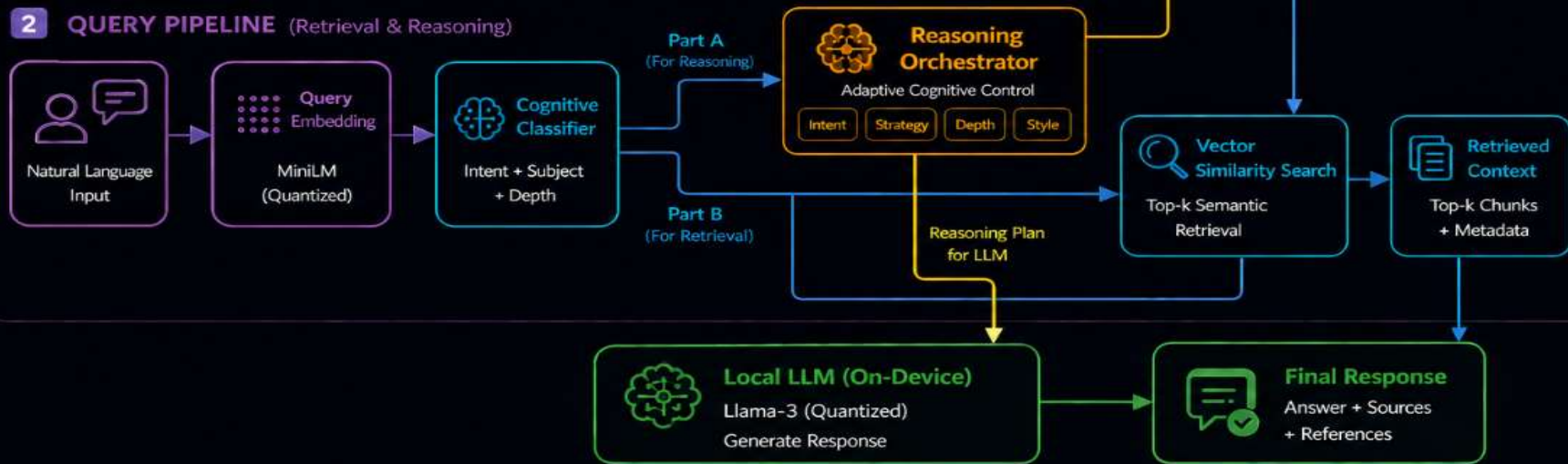
System Flow Diagram

Team: **Agile&Empathy**

1 INDEX PIPELINE (Data Ingestion & Semantic Indexing)



2 QUERY PIPELINE (Retrieval & Reasoning)



Indexing Pipeline
Data Ingestion & Indexing



Query Pipeline
User Interaction



Cognitive Layer
Understanding & Control



Reasoning Layer
Decision & Strategy



Retrieval Layer
Search & Context



Generation Layer
Response Generation

⚙️ Background Processing (Resource-Aware)

- Runs only during charging, idle, or screen-off
- Processes documents paragraph by paragraph
- Generates embeddings for each sentence
- Groups similar content into coherent chunks
- Merges related chunks across the document (e.g., OS topics)

🗄️ Dual Memory Architecture

Vector Database:

Stores only embeddings + chunk_id (fast semantic search)

Metadata Layer:

Stores full context, page numbers, file paths, references

🔍 Hybrid Cognitive Classifier

- Converts query into embedding vector
- Classifies Intent → Compare / Summarize / Explain
- Extracts Subjects → e.g. Transformer, CNN
- Prevents irrelevant retrieval (e.g. cats & dogs when comparing architectures)

🧠 Reasoning Orchestrator (Cognitive Core)

- Assesses ambiguity, complexity & depth needed
- Strategy → Shallow RAG / Deep reasoning / Comparison
- Decides response style (concise / educational / step-by-step)

Retrieval-Augmented Generation

Team: **Agile&Empathy**

RAG FLOW



✦ Key Advantages



Cross-Modal Search

Text query → returns diagrams, screenshots, or audio clips



Progressive Reasoning

Fast shallow answer first; deeper analysis loads async



Full Explainability

Every response cites exact source: file, page, line range

Vector Similarity Search

Cosine similarity over dense embeddings — fast ANN retrieval on-device

Temporary Semantic Graph

Built on-demand from retrieved chunks; discarded after response generation

Metadata Hash Lookup

Chunk ID → hash fn → exact file path, page, paragraph, modality type

Technology Stack

Team: **Agile&Empathy**

📄 EMBEDDING MODEL

MiniLM (INT8 Quantized)

👁️ VISION ENCODER

MobileCLIP

🔊 SPEECH TRANSCRIPTION

Faster-Whisper Tiny

📁 VECTOR DATABASE

LanceDB

📁 METADATA STORAGE

SQLite

📄 COGNITIVE CLASSIFIER

MiniLM + Rule-Based Parsing

⚙️ REASONING ORCHESTRATOR

Custom Lightweight Cognitive Engine

📄 LOCAL LLM

llama-3 Quantized

📱 ON-DEVICE INFERENCE

NNAPI (Android) • CoreML (iOS)

📄 BACKGROUND SCHEDULING

WorkManager

📄 OCR ENGINE

PaddleOCR

🔧 DEVELOPMENT STACK

Python



FULLY OFFLINE



EDGE-OPTIMIZED



12 COMPONENTS



0 CLOUD DEPENDENCIES



90–95%

Cost Reduction

Compared to combined cost of AI tools, cloud storage, and productivity apps



~8

Hours Saved / Week

Instant semantic retrieval across all personal knowledge sources



₹2.4B+

Market Opportunity

Privacy-first, offline AI tools (growing edge AI segment)

User Cost Comparison (Per Month)

Category	Traditional Tools	Our System
AI Tools	₹1600	₹0
Storage	₹300	₹50
Productivity Apps	₹1000	₹0
Transcription	₹700	₹0
Total	₹3600 / month	₹99–₹199 / month



User Savings: Up to 90–95%

Revenue Model (Tiered Pricing)



Free Tier

Limited usage



₹49 / month

Students



₹99 / month

Regular users



₹199 / month

Professionals

Break-Even Analysis

- Avg revenue per user ₹100
- Avg cost per user ₹10
- Profit per user ₹90
- Fixed monthly cost ₹1,00,000



Break-even users

≈ 1,100 users

Break-even = Fixed Cost / Profit per user
= 100000 / 90 ≈ 1,100 users

Profit Scaling (Conceptual)



User Growth Projection



Business Advantages



Fully offline AI (privacy-first)



Minimal server dependency → low cost



Multimodal intelligence (text, image, audio, video)



Faster than cloud-based workflows



Summary: Positioned as a Personal Cognitive AI Platform



Massive cost advantage for users



High-margin subscription model



Low infrastructure cost



Strong scalability potential

Future Impact & Vision

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EMPOWERING INDIVIDUALS & PROFESSIONALS

Students & Researchers

Instant semantic search across years of notes, papers & lectures with deep conceptual comparison

Knowledge Workers & Professionals

Personal second brain — remembers everything, enables smarter decisions, saves hundreds of hours yearly

Elderly & Lifelong Learners

Combats cognitive overload, supports memory, enables easy access to medical records & personal history

Creators & Entrepreneurs

Connects ideas across documents, images & recordings to spark innovation and faster execution

BROADER SOCIETAL IMPACT

 Democratizes advanced AI on everyday smartphones

 Reduces dependency on cloud & big tech for personal knowledge

 Digital longevity — intelligently preserves personal data for decades

 Future-ready for AR glasses, wearables & ambient computing

Conclusion

CogniMem is more than an AI tool — it is the foundation of a new era of personal intelligence. A truly private, adaptive cognitive companion that grows with you and amplifies human potential while protecting what matters most: your data and your privacy.

Thank You

We are excited to build the future of personal intelligence.

Q&A • Team: Agile & Empathy